VERIFAI: MEASURING AI WORKPLACE READINESS THROUGH REALISTIC JUDGEMENT TASKS

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ABSTRACT

Employers are adopting AI faster than graduates can be assessed on using it well. Tool knowledge is easy to test and easy to forget; what decides whether a junior hire is trustworthy with AI is judgement: catching a hallucinated citation, knowing when not to automate a decision, and spotting a privacy or bias risk before it reaches a client. VerifAI is a Flutter application that measures AI workplace readiness through realistic, employer-style work. Learners find the flaw in an AI output, correct it, and justify the decision. Every graded attempt feeds a deterministic, server-side readiness score across four South African industry tracks, producing a 0 to 100 readiness profile, a tier, and an evidence trail a recruiter can inspect.

INTRODUCTION

No CV states whether a graduate can be trusted with AI, and a multiple-choice quiz measures recall rather than judgement. The gap is the product: an assessment that behaves like the work itself.

The app runs a daily loop: learn a concept, practise a flawed scenario, be assessed under exam-style conditions, see the evidence, and keep the habit going. Content covers **Finance**, **Legal**, **IT and Software**, and **Business Consulting**, with 144 authored scenario tasks and 576 answer options.

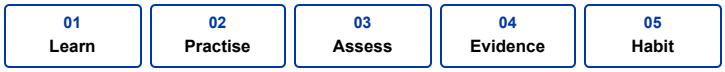


Figure 1: The VerifAI learner loop.

ASSESSMENT ENGINE

Verification. Either four-option choice, or a numbered AI output where the learner taps the flawed lines, types the correction, then the reason. Line-text scores 30 flaw, 40 fix, 30 reason, and needs 70 with a non-zero flaw *and* fix, so a plausible explanation cannot pass a task whose point is the correction.

AI Judgement. Decision (use AI, use with review, do not use) plus best-of-three reason. Decision scores 50, 25 or 0 for correct, neighbouring or opposite; reason scores 50 or 0. Pass 50.

Risk and Governance. Proceed or do not proceed, plus best-of-three reason, 50 and 0 each. Pass 50.

Interview Practice. An ungraded per-track question bank that feeds 10% of readiness as coverage.

Scoring is deterministic and server-side. The payload never contains the answer key, and no LLM grades a scored task, so the same answer always earns the same mark.

READINESS MODEL

Readiness is a weighted mean over the components a track actually offers. Each skill score is **coverage**: passed tasks divided by available tasks. A four-task skill therefore fills 25, 50, 75 then 100% as each task is passed, and never decays between sessions.

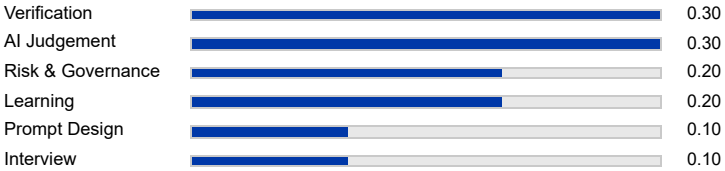


Figure 2: Readiness weights, fixed per track.

Novice	Developing	Proficient	Advanced
0 to 49	50 to 69	70 to 84	85 to 100

Tiers are derived from the live score, so they move with evidence.

SYSTEM DESIGN

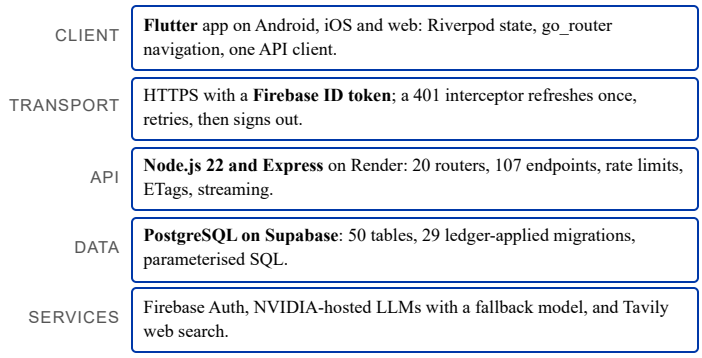


Figure 3: One API and one database behind three clients.

A submission is handled in one transaction: score the answer, write the attempt and its analytics row, recompute coverage and readiness if it moved, then award XP, tokens, streak and achievements. The app never computes a score.

IMPLEMENTATION

- **Scale:** 34.5k lines (12.5k backend, 21.9k Flutter), 40 screens across 22 feature modules.
- **Navigation:** five hubs, Home, Learn, Practice, Progress and Profile, with every other screen pushed from the hub that owns it.
- **Content:** 5 lessons with 248 sections, 4 timed capstones over 100 questions, 16 industry cases, 9 rewards and 13 achievements.
- **AI:** Verification Studio reviews pasted text, documents, links and citations; Vera and the in-task assistant stream over server-sent events and abort cleanly when the client disconnects.
- **Accessibility:** light, dark and system themes, bundled fonts, OS text scaling, and semantic labels on custom-drawn scores.

RESULTS

144	241	107	4
TASKS	TESTS	ENDPOINTS	TRACKS

The full loop is working end to end against the deployed API. A submission moves readiness, XP and tier immediately; feedback is stored and reopens from history or a deep link; and the same build ships as a web app and an installable APK. Continuous integration runs 186 backend and 55 Flutter tests, static analysis, a dependency audit and a secret scan on every push.

An internal audit closed 24 findings across three rounds, including fixed per-track readiness denominators, a shared rate-limit store in PostgreSQL, off-thread document parsing, and cached static payloads with ETag revalidation.

CONCLUSION AND FUTURE WORK

VerifAI turns a vague hiring question, whether a graduate can be trusted with AI, into a measurable readiness score backed by evidence. The engineering that makes it credible is already in place: one authoritative scoring pipeline, a schema that evolves through reviewed migrations, streaming AI features that degrade gracefully, and server-side authorisation for every privileged action.

Outstanding work is scheduled rather than open: a testing phase to 30 October covering regression and a device matrix, Prompt Design content (scaffolded but not authored), a Workshop learner screen (API and moderation already complete), an embedded-Postgres integration suite, and row-level security hardening on the hosted database.